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Sri Mats Engineering

Class Test - 01

T. Bhanu Teja

1a2472259

MLA0301

Reinforcement learning.

- ① Define Reinforcement Learning (RL). Explain its main objective.

Def:- Reinforcement learning (RL) is a branch of machine learning in which an agent learns by interacting with an environment. The agent performs actions, receives rewards or penalties, and learns the best policy through trial and error.

Main objective:-

The main objective of RL is to learn an optimal policy that maximizes the cumulative (long-term) reward while interacting with the environment.

- ② What is a Markov Decision Process (MDP)? List its components.

A Markov Decision process (MDP) is a mathematical framework used to model sequential decision-making problems where outcomes depend on the current state and action.

Components of MDP

- 1) State (s)
- 2) Action (A)
- 3) Reward (R)
- 4) Transition Probability
- 5) Discount factor (γ)
- 6) Policy (π)

3. State the Bellman Equation for the state-value function.

The Bellman Expectation Equation for the state-value function is:

$$V(s) = R + \gamma V(s')$$

Where,

$V(s)$ = value ; R = Immediate Reward ; γ = Discount factor.

$V(s')$ = value of next State.

The Bellman Equation updates the value of a state using immediate and future rewards.

4. Differentiate btwn Exploration and Exploitation.

Exploration :

- Tries new actions
- Gains new knowledge
- may receive lower immediate reward
- helps discover better strategies.

Exploitation :

- Chooses the best-known action
- uses existing knowledge
- usually gives higher immediate reward
- Maximizes Current reward.

5. What is the Multi-Armed Bandit problem?

The Multi-Armed Bandit Problem is a RL problem where an agent chooses among several actions (slot machines or bandits) with unknown rewards. The goal is to maximize the total reward by balancing exploration and exploitation (using the best-known action).

What is Dynamic Programming (DP)? mention its applications in RL.

Dynamic Programming (DP) is a method used to solve Markov Decision processes when the complete environment model is known.

Applications

- * Policy Evaluation
- * Policy Improvement
- * Policy Iteration
- * Value Iteration
- * Finding Optimal Policy

⑦ What is Temporal Difference (TD) Learning? write the TD(0) update equation.

Temporal difference (TD) Learning is a model-free RL method that updates value estimates using the difference between predicted and actual reward.

TD(0) update equation.

$$V(s) = V(s) + \alpha [R + \gamma V(s') - V(s)]$$

Where :- α = Learning Rate

γ = Discount factor

R = Reward

$V(s')$ = Next state value.

⑧ Differentiate btwn SARSA and Q-Learning?

SARSA

Q-Learning

→ off-policy algorithm

→ on-policy algorithm

→ uses the next selected action.

→ Learns according to current policy

→ more conservative

→ Uses the maximum future action.

→ Learns the optimal policy

→ more aggressive

⑨ what is a Deep Q-Network (DQN)? Mention its advantages over tabular Q-learning.

A Deep Q-network (DQN) is a RL algorithm that uses a Deep neural network to approximate Q-values instead of storing them in a table.

Advantages over Tabular Q-learning:

- * handles large state spaces.
- * works with high dimensional inputs (images, videos).
- * generalizes to unseen states.
- * more scalable.
- * learns complex decision-making tasks.
- * suitable for real-world applications.

⑩ what is Double Deep Q-Network (DDQN)? Why is it preferred over DQN?

A DDQN is an improved version of DQN that uses two networks (an online network and a target network) to reduce overestimation of Q-values.

Why DDQN is preferred:

- * Reduces overestimation bias.
- * produces more accurate Q-value estimates.
- * Improves learning stability.
- * achieves better performance.
- * provides faster convergence.
- * makes more reliable decisions in complex environments.